Yohan Abeysinghe

+1 (227) 205-7077 abeysingheyohan@gmail.com Website GitHub LinkedIn

EDUCATION

University of Maryland, College Park, USA **Aug 2025 – Present**
Ph.D. in Electrical and Computer Engineering *GPA: 3.85* *Advisor: Prof. Dinesh Manocha*

University of Moratuwa, Sri Lanka **Jan 2020 – Jul 2024**
B.Sc. (Hons) in Electronics and Telecommunication Engineering *GPA: 3.89* *First Class Honours*

Maliyadeva College, Kurunegala, Sri Lanka **Dec 2018**
GCE Advanced Level: Mathematics, Physics, Chemistry *3 A's* *Z-score: 2.78* *Island Rank: 24*

PUBLICATIONS

- Gershom Seneviratne, **Yohan Abeysinghe**, Jianyu An, Vaibhav Shende, Rahul Kumar, Dinesh Manocha. "VEGA: Learning Navigation VLAs from In-the-Wild Egocentric Video with Geometric Trajectory Supervision." *arXiv preprint arXiv:2606.18426*, 2026. [Paper] [Project] [Video]
- **Yohan Abeysinghe**, Muhammad Akhtar Munir, Sanoojan Baliah, Ron Sarafian, Fahad Shahbaz Khan, Yinon Rudich, Salman Khan. "Synergistic Neural Forecasting of Air Pollution with Stochastic Sampling." *npj Clean Air*, vol. 2, article 36, 2026. [Paper] [arXiv] [Code]
- Sanjoy Chowdhury, Mohamed Elmoghany, **Yohan Abeysinghe**, Junjie Fei, Sayan Nag, Salman Khan, Mohamed Elhoseiny, Dinesh Manocha. "MAGNET: A Multi-agent Framework for Finding Audio-Visual Needles by Reasoning over Multi-Video Haystacks." In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. [Paper] [Project] [Code]
- Sanoojan Baliah, **Yohan Abeysinghe**, Rusiru Thushara, Khan Muhammad, Abhinav Dhall, Karthik Nandakumar, Muhammad Haris Khan. "VFace: A Training-Free Approach for Diffusion-Based Video Face Swapping." In *Winter Conference on Applications of Computer Vision (WACV)*, 2026. [Paper] [Code]

RESEARCH EXPERIENCE

University of Maryland, College Park, USA **Aug 2025 – Present**
Doctoral Researcher *Advisor: Prof. Dinesh Manocha*

- Trained and deployed VEGA vision-language-action models for robot navigation, translating policies learned from geometrically supervised egocentric video into real-robot execution.
- Currently developing a self-supervised variant of VEGA for learning robot navigation policies.
- Developing runtime steering for navigation policies using Euclidean signed distance fields and dynamic-obstacle velocity estimates; evaluating goal reaching and obstacle avoidance through real-robot experiments and controlled ablations.

Mohamed bin Zayed University of Artificial Intelligence, UAE **July 2024 – July 2025**
Research Associate *Advisor: Dr. Salman Khan*

- Developed SynCast, a spatiotemporal air-pollution forecasting framework combining a pretrained weather transformer, LoRA-based regional adaptation, and diffusion-based stochastic refinement.
- The model reduced 24-hour $PM_{2.5}$ forecasting RMSE by approximately 18–39% against Aurora and AirCast across evaluated resolutions in the Middle East and North Africa.
- First-authored the resulting paper, published in *npj Clean Air* (2026).

University of Moratuwa, Sri Lanka **Sept 2023 – June 2024**
Undergraduate Thesis Research Student *Advisor: Dr. Chamira Edusooriya*

- Developed attention-based temporal convolutional models for motor imagery and P300 target/non-target EEG classification.
- Applied Grad-CAM to EEG scalograms to examine time-frequency regions contributing to classification decisions.
- Built an ear-EEG acquisition hardware prototype, including a fabricated printed circuit board and a 3D-printed earpiece.

University of Sydney, Australia **Jan 2023 – Sept 2023**
Research Assistant - Internship *Advisor: Dr. Anusha Vithana*

- Implemented ML-based structure prediction pipelines and compared intermediate model outputs against physics-based simulations to evaluate learned representations and real-world dynamical behavior.

SELECTED HONORS AND AWARDS

Bronze Medal, International Mathematical Olympiad (IMO) 2017
Honourable Mention, International Mathematical Olympiad (IMO) 2014, 2015, 2016
Bronze Medal, Asian Pacific Mathematics Olympiad (APMO) 2018
Honourable Mention, International Mathematics Competition for University Students 2022
Honourable Mention, Asian Physics Olympiad (APhO) 2019
Gold Medal, Sri Lankan Physics Olympiad 2019

OTHER SELECTED PROJECTS

Vision-based Object Recognition and Task Solving Robot **Feb 2022 – Aug 2022**

- Built a vision-guided robot navigation pipeline in Webots using camera inputs and OpenCV/C++, connecting visual perception outputs to autonomous task execution.

AHEAD Intelligent Camera Project **Dec 2022 – Mar 2023**

- Implemented real-time YOLO-based object detection and face recognition on an embedded Axis P1341 camera, optimizing vision inference for resource-constrained deployment.

Protein Folding: Free Energy Landscape with ML **2023**

- Implementing RoseTTAFold protein folding algorithm and taking the intermediate structures that a protein goes through while the algorithm is predicting the final structure.
- Using GROMACS Molecular Dynamics simulations to identify the folding pathways of a protein. Checking the similarities between the intermediate stages from two methods to identify how much of real world physics is learned by folding algorithms.

Project Eyeguard **Mar 2021 - Feb 2022**

- Developing a device to minimize the eye defects that can arise from long-term screen usage, by using cameras to track the eye and the blinking rates, and sending feedback to the users to take actions accordingly.

SKILLS

- Python
- PyTorch
- TensorFlow
- MATLAB
- ROS

RELEVANT COURSEWORK

Computer Vision: EN4553 Machine Vision (A+), EN4583 Advances in Machine Vision (A+), EN4573 Pattern Recognition and Machine Intelligence(A+)
Signal Processing: ENEE620 Random Processes in Communication and Control (A-), ENEE627 Information Theory (A-), ENEE630 Advanced Digital Signal Processing (A)
Mathematics: MA2023 Calculus (A+), MA2033 Linear Algebra (A), MA4013 Linear Models and Multivariate Statistics (A), MA2013 Differential Equations (A)

LEADERSHIP/EXTRACURRICULARS

President: Mathematics Society, University of Moratuwa 2021
Student Representative: Department of Electronic and Telecommunication, University of Moratuwa 2022
Committee Member: Electronics Club, University of Moratuwa 2023
Sergeant at Arms: Gavel Club, University of Moratuwa 2022
People Management Team: AIESEC in Colombo South 2021
Prefect: Maliyadeva College 2018

REFERENCES

Prof. Dinesh Manocha
Distinguished University Professor
Department of Computer Science
University of Maryland
dmanocha@umd.edu

Prof. Salman Khan
Professor
Computer Vision Department
Mohamed bin Zayed University of Artificial Intelligence
salman.khan@mbzuai.ac.ae